

Amogh Athimamula

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EDUCATION

Northeastern University - Khoury College of Computer Science

Boston, MA

Bachelor of Science in Computer Science, Minor in Business Admin. - **3.98 GPA**

August 2024 – May 2028

Relevant Coursework: Object Oriented Design, Databases, Data Structures + Algorithms, Fundamentals of Computer Science I/II, Cloud Computing, Foundations of Data Science, Intro to Cybersecurity

TECHNICAL SKILLS

AI/ML & Dev Tools: PyTorch, Gemini API, OpenAI Codex, Claude Code, Cursor

Languages: Python, Java, Go, C++, C#, SQL, JavaScript, TypeScript, HTML/CSS, Verilog HDL

Frameworks & Libraries: React Native, React, Node.js, Express.js, FastAPI, Expo, Prisma, Material-UI, Unity

Databases: PostgreSQL, MySQL, SQLite, Redis, Supabase

Tools & DevOps: Git, Docker, AWS (Lambda, S3, ECS, ASG, CloudWatch), Azure, Jenkins, Jira

EXPERIENCE

Software Engineer Intern - Pethealth Storefront

June 2026 – Present

Chewy

Boston, MA

- Spearheaded AI-assisted migration of legacy **WebView** screens to native bottom sheets, taking initiative beyond the original single-screen scope to build reusable automation infrastructure – migrated **10+ screens** from a zero baseline while cutting per-screen migration time from a **2-week sprint to 1-2 days (90% reduction)**
- Rewired vet diet authorization logic connecting **pethealth-b** and **RxP**, extending the request contract with **4 new fields** (quantityType, quantityValue, orderId, orderItemId) and migrating decision authority from a local status check to RxP's dedicated prescription-needed endpoint, consolidating auth logic into a single integration point

Software Engineer Co-op - Fixed Income Platform

January 2026 – June 2026

Wellington Management

Boston, MA

- Shipped **CLI disaster recovery tool** supporting **150+ stacks** with individual stack-level targeting, automating cross-region **AWS** failover across pre, active, and post-recovery phases, orchestrating **ASG/ECS** infrastructure and DNS switching with live deviation checks, reducing manual recovery effort by **75%**
- Replaced manual email triage for **20 release engineers** by engineering a full-stack alert dashboard surfacing **2,000+** categorized alerts via **AWS Lambda, Python**, and optimized **SQL**, auto-flagging critical incidents
- Eliminated third-party BI tool dependency by engineering a scheduled pipeline with Python and SQL to generate interactive HTML dashboards tracking SLA compliance, surfacing breach patterns across Fixed Income systems facilitating billions in daily trading and collaborating with engineering teams to drive remediation
- Refactored legacy health check utility serving all portfolio applications, optimizing **AWS Lambda** execution to reduce runtime by **30%** through improved resource allocation and code efficiency

Software Engineer

September 2025 – Present

Generate - Product Development Studio

Boston, MA

- Building **Toggo**, a full-stack group travel planning mobile app with a **6-person team** using **Expo, React Native, Go, PostgreSQL, Redis, WebSockets**, and **AWS S3**, following weekly **Agile** sprints for a client-driven MVP
- Architected and implemented **Expo push notification service** using **Go SDK**, supporting **4 notification types** with bulk multi-user delivery, device token persistence, and mock-tested external service integration
- Built end-to-end **trip pitch feature** with **5 REST endpoints** and pagination, **S3 audio storage** via presigned URLs, and automated push notifications triggering on pitch creation for all group members

PROJECTS

S.H.I.E.L.D. Threat Analysis Platform | Python, PyTorch, FastAPI, SQLAlchemy, Gemini API, SQLite

- Engineered **AI threat monitoring pipeline** processing **300+** daily news articles via **NewsAPI** and **Gemini API**, with automated deduplication, batch scheduling, and 1–10 severity scoring
- Trained a **PyTorch** text classification model on labeled threat articles to pre-screen and prioritize ingestion, reducing Gemini API calls by **35%** while maintaining classification accuracy above **90%**
- Exposed **8+ REST endpoints** via **FastAPI** supporting threat retrieval, severity filtering, and human review workflows, with **APScheduler** driving hourly automated processing

FastBook – FPGA Order Execution Engine | Verilog HDL, C++, Vivado, Python

- Designed and simulated a **hardware order book** in **Verilog HDL** using **Xilinx Vivado**, implementing price-time priority matching logic for limit and market orders with **sub-100ns match latency** in simulation
- Built a **C++ order injector** to drive the testbench with randomized and adversarial order streams, validating correctness of fill logic, partial fills, and order cancellation across **10,000+** simulated events
- Instrumented throughput and timing analysis in **Python** to benchmark match engine performance, identifying and resolving a pipeline stall that improved simulated throughput by **22%**